

Experimental study of bond-slip of GFRP bars in concrete under sustained loads



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ABSTRACT

The structural behaviour of reinforced concrete (RC) elements depends heavily on the bond performance between the concrete and the reinforcing material. Bond behaviour under short-term testing has been extensively analysed for steel reinforcement and many studies have been carried out for fibre reinforced polymer (FRP) reinforcement. However, there has only been limited investigation of the long-term effects of this interaction. Several factors can affect the long-term bond behaviour of these elements, the most important being bond length and the immediate and time-dependent properties of reinforcement and concrete (concrete grade, creep, shrinkage and stiffness). This time-dependent behaviour is likely to cause changes and redistributions in bond stresses not properly considered in the limited existing literature. In this experimental study, the bond performance of GFRP RC under sustained load is investigated through pull-out tests. A total of 12 pull-out specimens were tested for a period of between 90 and 130 days. Two concrete strengths (35 MPa and 50 MPa), two bond lengths (5 and 10 times the diameter of the reinforcing bar) and two reinforcing materials (glass fibre reinforced polymer (GFRP) and steel) were used. Experimental results regarding immediate and time-dependent slip are presented and analysed here. In addition, some specimens were instrumented, with internal strain gauges in the reinforcing bar to provide data on the reinforcement strain, thus allowing the distribution of bond stresses and their evolution during sustained loading to be also presented and analysed.

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1. Introduction

Fibre reinforced polymer (FRP) bars are increasingly being used as an alternative to steel reinforcement for reinforced concrete (RC) elements in corrosive environments or when the effects of electromagnetic fields may be present [1–3]. This has led to an increasing interest in the knowledge of properties and the study of different aspects of behaviour of FRP RC structures [4–14]. The viability of these recently introduced materials largely depends on the effectiveness of the bond between the FRP bar and the concrete. In this regard, pull-out tests are probably one of the tests most extensively used to characterize the behaviour of the interface between the reinforcement and the concrete by means of bond-slip response. In the last two decades, considerable experimental research has gone into investigating the short-term response of FRP-to-concrete interfaces [15–24], with it being generally concluded that the pull-out mechanism of the many existing types

of FRP reinforcement differs from that of deformed steel bars and is dependent on even more parameters [20,25].

When analysing time effects on the behaviour of RC elements, concrete creep and shrinkage play a crucial role [26–32]. While creep is associated with sustained stresses, shrinkage may be assumed to be independent of load, but both cause long-term deformations in concrete. For common steel RC no additional long-term effects need to be considered because steel undergoes neither shrinkage nor creep. Few studies have focused on the possible effect of creep in FRP reinforcement. An experimental programme with regard to such an effect in aggressive environments was presented in Ref. [33]. The programme consisted of testing twenty GFRP bars in tension for 417 days at two loading levels (25% and 38% of the ultimate bar tensile strength). The results showed that creep strain in the GFRP bars was less than 5% of the initial strain value. It was also concluded that long-term loading had a minimal effect on the elastic modulus, whilst the effect on residual strength was more dependent on the environmental conditions. A more recent programme was presented in Ref. [34], where GFRP pultruded laminates were tested under a tensile

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sustained load for 500 days. The authors observed that the largest increments in longitudinal stresses due to long-term testing were less than 2%. The components (matrix and reinforcement or polyester resin and glass fibres) were tested separately to check their individual long-term responses. A significant increase in longitudinal strain was observed for the polyester resin while only a negligible increase was obtained for the glass fibres.

The literature dealing with time effect on the bond behaviour of FRP reinforcement and concrete is also limited in extent [35–38]. An experimental study of long-term behaviour of CFRP externally reinforced concrete elements covering three bond lengths and two loading levels was presented in Ref. [35], where the evolution of reinforcement axial stresses and bond stresses under constant tensile load was analysed over 900 days of testing. Results showed a redistribution of bond stresses along the anchorage due to creep deformations at the interface. In respect of internal reinforcement, degradation of the bond between FRP bars and concrete under sustained load was analysed in Ref. [36] through a pull-out testing procedure. The experimental work included four different reinforcing materials as well as different environmental conditions, with pull-out specimens being loaded for one year at different loading levels. Results showed an increase in slip with time, with this increase being dependent on the bar surface treatment and the level of sustained load. A more recent experimental study of the bond behaviour of internal GFRP reinforcement and concrete under sustained loading was presented in Ref. [37]. The pull-out specimens, which combined three different bar diameters, were pre-loaded until one millimetre of slip was observed at the unloaded end. Following this, the specimens were loaded until slip stabilization. The tests were considered to be acceptable if no increase in slip was observed after 2000 h. To simulate accelerated long-term testing, the tests were conducted at 60 °C, and a design value of bond strength was proposed by the authors based on the experimental data. The long-term effect on steel and GFRP internally reinforced concrete members under tension has been analysed in a study made by the present authors [38]. The experimental programme included three different concrete strengths and two different reinforcing materials. The evolution of reinforcement strains was monitored, and slips and bond stresses were analysed. Results showed that long-term testing caused a reduction in mean bond stress of about 28%, which highlights the importance of analysing and understanding bond behaviour in the long term. Since limited literature exists on this issue, an experimental long-term bond test programme is needed.

This paper presents the results of an experimental campaign consisting of twelve pull-out specimens tested under sustained axial load for a period between 90 and 130 days. The programme included two concrete strengths, two reinforcing materials and two bond lengths. The sustained load level was set at 15% of the ultimate capacity of the GFRP reinforcing bar to ensure it corresponded to the service load range. Experimental results in terms of immediate and time-dependent slip are presented and analysed. Some specimens were instrumented, with internal strain gauges in the reinforcing bar, to provide data on the reinforcement strain, thus allowing the distribution of bond stresses and their evolution during sustained loading to be analysed.

2. Experimental programme

2.1. Test matrix

The experimental programme was aimed at studying the effect of concrete strength, reinforcing material and bond length on the bond response of pull-out tests under sustained loading. Two different target concrete compressive strengths (35 and 50 MPa),

two different bond lengths (equal to $5d_r$ and $10d_r$, with d_r being the nominal diameter of the reinforcing bar) and two reinforcing materials (GFRP and steel) were considered. So that axial stiffness (EA) would be similar in all tests, the reinforcement consisted of either a single 16 mm diameter GFRP bar or a single 10 mm diameter steel bar. The combination of these variables gave a total of eight specimens. So that the internal distribution of bond stresses could be analysed, four additional specimens were manufactured that included internal instrumentation of the GFRP reinforcing bar. Thus, the test matrix consisted of twelve pull-out specimens divided into two groups of six specimens each according to concrete grade.

Based on this description, the tested elements are identifiable by the formula CxBRi, with Cx standing for the type of concrete (C1 = 35 MPa, C2 = 50 MPa), B for the bond length ($S = 5d_r$, $L = 10d_r$), R for the type of reinforcement ($F = \text{GFRP}$, $S = \text{Steel}$), and “i” for the identification of specimens with internally instrumented reinforcement. The test matrix is summarized in Table 1.

All the specimens were cubic (200 mm sides) with a 600 mm bar located in the middle. Before casting, the bond length was appropriately marked, and a plastic tube was positioned to prevent contact along the remaining bar length. Steel housings were glued to the GFRP bars so that the pull-out load could be applied without damaging the bars.

2.2. Material properties

Ready-mixed concrete was used to cast the specimens. Compressive strength was determined at the time of loading (35 days after casting) by a standard cylinder test (150 × 300 mm) according to UNE 12390-3 [39]. The average values of the mechanical properties are summarized in Table 2.

In order to determine the characteristics of the GFRP and steel reinforcement, three samples for each material were tested under tension according to UNE ISO 15630-1:2011 [40] and ACI 44.3R-12 [41] respectively. The average values of the mechanical properties of the reinforcing bars are given in Table 3.

2.3. Test set-up

Three frames with a double lever system were used to apply a constant tensile load on the pull-out specimens (see Fig. 1). The amplification factor of the mechanical system was 11. In each frame, two pull-out specimens with the same reinforcing bar and concrete grade but of different bond length were connected in series (i.e. specimen C1SS was connected in series with specimen C1LS), which meant that the same load was applied to both

Table 1
Test matrix.

Specimen	Concrete	Bond length, l_b (mm)	Reinforcement	Reinforcement internal instrumentation
C1SS	C1	50	Steel	No
C1SF	C1	80	GFRP	No
C1SFi	C1	80	GFRP	Yes
C1LS	C1	100	Steel	No
C1LF	C1	160	GFRP	No
C1LFi	C1	160	GFRP	Yes
C2SS	C2	50	Steel	No
C2SF	C2	80	GFRP	No
C2SFi	C2	80	GFRP	Yes
C2LS	C2	100	Steel	No
C2LF	C2	160	GFRP	No
C2LFi	C2	160	GFRP	Yes

Table 2
Mechanical properties of concrete.

Concrete	Compressive strength, f_c (MPa)	Elastic modulus, E_c (GPa)
C1	35.5	32.1
C2	48.5	35.3

Table 3
Mechanical properties of reinforcement.

Material	Elastic modulus, E_r (GPa)	Tensile/Yielding strength (MPa)	Axial stiffness, EA (MN)
GFRP	66.5 ± 2.5	1200 ± 61	13.5 ± 0.5
Steel	206 ± 5.0	520 ± 10	16.2 ± 0.4

specimens. A maximum load equal to 15% of the ultimate capacity of the GFRP bar was selected (see Table 4), in order to correspond to the service load range (ACI 440 1.R-06 [42] fixes a limit of 20% for GFRP bars). The sustained load applied to the steel reinforced specimens was adapted (by means of the bond perimeter) so as to obtain similar bond stresses to those of GFRP reinforced pull-outs.

To perform the test the load was applied progressively using $400 \times 400 \times 15$ mm (18.1 kg) steel plates, until the loads indicated in Table 4 were reached. These maximum loads were thereafter sustained for 90 days in the case of series C1 and 130 days in series C2.

To measure the slip at both the loaded and the unloaded end, two displacement transducers (HLS) were used for each pull-out specimen (see Fig. 2a). Four of the twelve pull-out specimens allowed monitoring of the strain distribution along the bar. These specimens were cast using specially manufactured internally strain gauged GFRP bars [38,43]. The location of the internal instrumentation is shown in Fig. 2b. In the case of the short bond length specimens, four and two strain gauges were placed along and outside the bond length respectively. For long bond length specimens, the number of gauges located along the bond length was doubled to eight.

Table 4
Maximum loads.

Specimens	P_{max} (kN)
C1SS, C1LS	10.5
C1SF, C1LF	29.5
C1SFi, C1LFi	29.5
C2SS, C2LS	10.5
C2SF, C2LF	29.5
C2SFi, C2LFi	29.5

2.4. Shrinkage and creep

The creep coefficient for each series was determined in accordance with ASTM C5 12-02 [44]. Two concrete cylinders (150 mm in diameter and 450 mm in length) with embedded strain gauges were stacked in a loading frame. The cylinders were loaded at the same time as the pull-out specimens. An additional 200 mm cubic concrete specimen was instrumented and left unloaded to determine free shrinkage strain.

Temperature and humidity were also recorded. An average temperature and relative humidity of 16 ± 2.5 °C and $40 \pm 23.5\%$ were registered for the C1 series. These registers changed to 27 ± 3.5 °C and $49 \pm 15\%$ for the C2 series. Recorded values are shown in Fig. 3.

The evolution of the shrinkage strains recorded for the two types of concrete is shown in Fig. 4a. The values of free shrinkage strains recorded at time t ($\epsilon_{sh}(t, t_0)$) are indicated in Table 5. As can be observed, from 10 to 90 days the shrinkage strain increased from $-46 \mu\epsilon$ to $-190 \mu\epsilon$ for C1 (35 MPa), and from $-36 \mu\epsilon$ to $-182 \mu\epsilon$ for C2 (50 MPa). At 90 days, the shrinkage strain of concrete C1 was 4.2% greater than that of concrete C2.

The experimental creep coefficient obtained from the strain measurements of the C1 and C2 series, calculated from the ratio of creep strain to instantaneous strain, is shown in Fig. 5. The creep strain at time t ($\epsilon_{\phi}(t)$) is obtained by subtracting the shrinkage strain, $\epsilon_{sh}(t, t_0)$ from the total strain $\epsilon_t(t, t_0)$ [30,38]. At the beginning of the test the creep effect increases considerably with time, but thereafter tends to increase slowly. The average values of experimental creep coefficient ($\phi(t, t_0)$) are indicated in Table 5. At 10 days after loading, the creep coefficient reads 1.00 and 0.74 for C1 and C2 respectively. From 10 to 90 days after loading, it increases from 1.00 to 1.88 for C1 and from 0.74 to 1.37 for C2, corresponding to increases of 88% and 85% respectively. The influence of concrete strength on creep coefficient is observable in these values, with the creep coefficient at 90 days of C1 being around 1.37 times greater than that of C2.

3. Test results

The results obtained in the experimental tests described in Section 2 are presented and discussed in the following. Particularly, the immediate value and the time-development of the relative slip measured at the bar–concrete interface are specifically considered. As stress distribution is not constant along the bond length in a pull-out test, an average bond stress is defined as:

$$\tau_{av} = \frac{P_{max}}{\pi d_r l_b} \quad (1)$$

where τ_{av} is the average bond stress applied to the interface, P_{max} is the maximum load applied to the bar, d_r is the bar diameter and l_b is the bond length.

For the test set-up, the pull-out specimens were loaded in pairs of equal concrete grade and with the same reinforcing bar, but

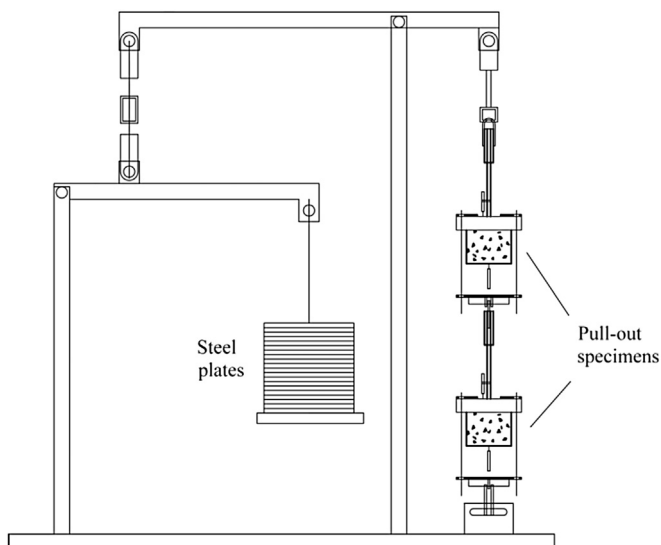


Fig. 1. Frame for long-term testing.

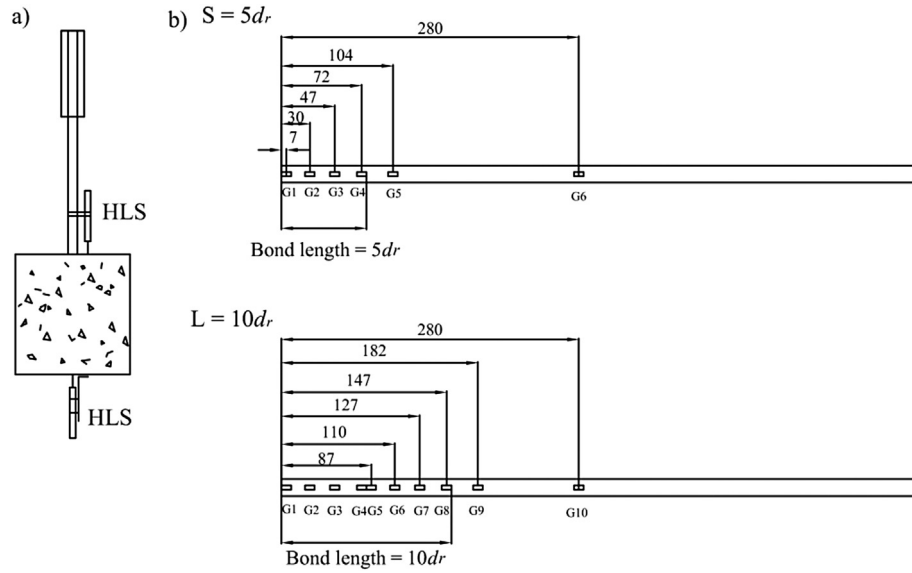


Fig. 2. a) External instrumentation of the pull-out specimens b) Internal instrumentation of the GFRP reinforcing bar (distance in mm).

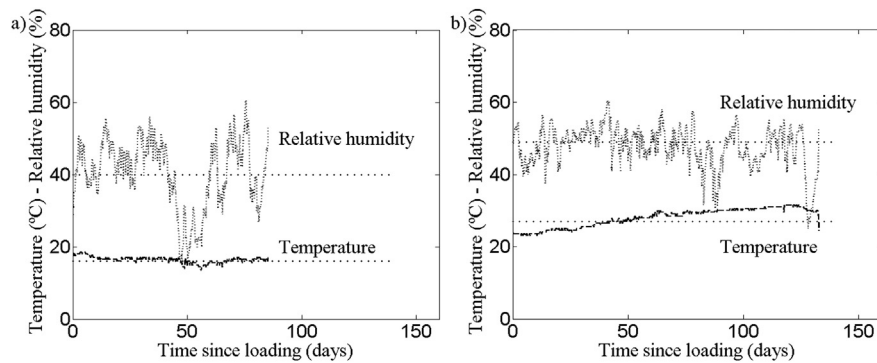


Fig. 3. Temperature and relative humidity registered in the laboratory during a) C1 tests and b) C2 tests.

different bond length (i.e. specimen C1SS and C1LS in series, specimens C1SF and C1LF in series, and so on). As a result, the average bond stress induced in short bond length specimens was twice that in long bond length specimens (see Table 6).

Experimental total slips versus time after loading are shown in Figs. 6 and 7. The total slip includes both the immediate slip due to instantaneous loading and the time-dependent slip due to sustained loading. As can be observed, the curves have a similar trend, with a larger immediate slip for short bond length specimens.

3.1. Immediate slip

Experimental data on the immediate response is given in Table 7 and analysed in this section. As can be seen values of registered slips are influenced by bond length, concrete strength and type of bar as has been reported in previous studies [19–24].

Irrespective of the type of bar and bond length, slips are larger for lower concrete strength which is a consequence of larger local deformability and damage in the concrete surrounding the bar.

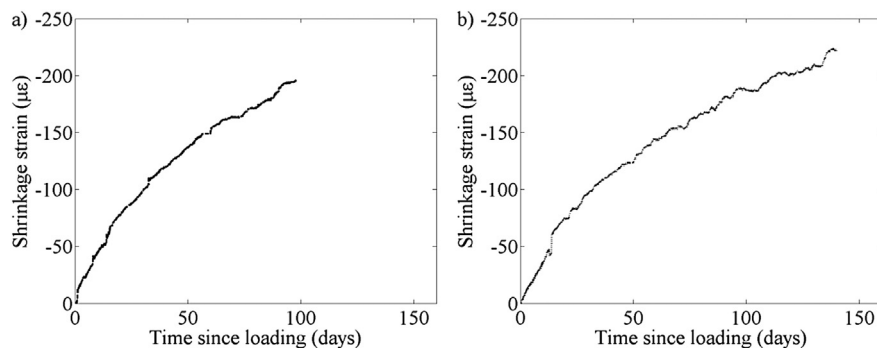


Fig. 4. Experimental free shrinkage strain of a) C1 series and b) C2 series.

Table 5

Experimental time-dependent concrete properties (average values) from the loading day.

Concrete	Time, t (days)	Creep coefficient, $\varphi(t, t_0)$	Shrinkage strain, $\epsilon_{sh}(t, t_0)$ ($\mu\epsilon$)
C1	10	1.00	–46
C1	50	1.57	–134
C1	90	1.88	–190
C2	10	0.74	–36
C2	50	1.19	–123
C2	90	1.37	–182
C2	130	1.57	–223

Likewise, slips are larger for specimens with lower bond lengths due to the higher bond stresses needed to transfer similar loads. It is also observed that the difference in slips between loaded and unloaded end is larger for the FRP bars as reported in Refs. [19,21]. This may be attributable to the higher deformability of the FRP bars jointly with a more effective mechanical interlock between bars and concrete due to the material and surface characteristics. Comparing similar specimens with different type of bar, slips are generally higher for the FRP bars, which indicates a lower slope in the bond stress-slip response. The only exception is for the slip at the unloaded end in specimens with short bond length and C1 concrete, which may be attributable to the combination of higher bar stiffness and damage introduced in the concrete for this short bond length. Globally, these differences in slips between both ends indicate a non-uniform distribution of stresses along the bonded length, which is more pronounced for larger bond lengths [23,24].

3.2. Time-dependent slip

For the sake of clarity and to make analysing the long-term experimental results easier, in Figs. 8 and 9 the immediate slip has been removed, so that only the experimental time-dependent slip (Δs) is presented. In addition, representative time-dependent slips at different times during long-term testing have been tabulated in Table 8.

Both Figs. 8 and 9 and Table 8 confirm the minimal effect of long-term testing on the slip at the unloaded end. As an example, time-dependent slip at 90 days (Δs_{90}) is below 0.012 mm for steel specimens and below 0.022 mm for GFRP specimens. Larger time-dependent slips are observed for the loaded end. An analysis of the evolution of these slips is presented below.

Experimental time-dependent slip results for C1 specimens are given in Fig. 8. As can be seen in the slopes of the curves, at the beginning of the sustained loading test the slip increases considerably over time, but thereafter tends to increase more slowly. Changeover takes place at around 60 days, when an average of

Table 6

Maximum applied loads and average bond stresses at the interface.

Specimen	P_{max} (kN)	Average bond stress, τ_{av} (MPa)
C1SS	10.5	6.71
C1SF	29.5	7.33
C1SFi	29.5	7.32
C1LS	10.5	3.35
C1LF	29.5	3.67
C1LFi	29.5	3.66
C2SS	10.5	6.71
C2SF	29.5	7.33
C2SFi	29.5	7.32
C2LS	10.5	3.35
C2LF	29.5	3.67
C2LFi	29.5	3.66

89.7% and 97.8% of the time-dependent slip at 90 days (Δs_{90}) is achieved for short and long bond length specimens respectively. Finally, the time-dependent slip at 90 days (Δs_{90}) for specimens with a long bond length is 1.94 times that of specimens with a short bond length.

Similar trends can be observed in C2 specimens (see Fig. 9 and Table 8). In this series, the sustained load was applied for a total of 130 days, making stabilization much clearer. In fact, based on the stabilization observed previously in this series, the C1 series was designed to consist of 90 days' testing. At 60 days an average of 95.5% and 95.9% of the time-dependent slip at 90 days (Δs_{90}) was achieved for short and long bond length specimens respectively. In this series, the time-dependent slip at 90 days (Δs_{90}) for specimens with long bond length was 1.49 times that of short bond length specimens. According to these results, the time-dependent slip at 90 days (Δs_{90}) in the C2 series is around 23.3% lower than in the C1 series. This difference could be attributable to the mechanical properties of concrete (immediate and time-dependent), where lower concrete strength produces higher time-dependent slips, as would be expected.

A summary of experimental time-dependent slips at 60 and 90 days after loading (Δs_{60} and Δs_{90}) is shown in Table 8. Additionally, the time-dependent slip at 130 days (Δs_{130}) is also reported for the C2 series.

3.3. Bond stresses – distribution and evolution

Experimental results show an increase in pull-out slips due to sustained loading, attributable to bond deterioration. There are no means of directly registering the bond taking place at an interface, but bond can be estimated from strain distribution along the bond length, as shown in previous works [35,38,43].

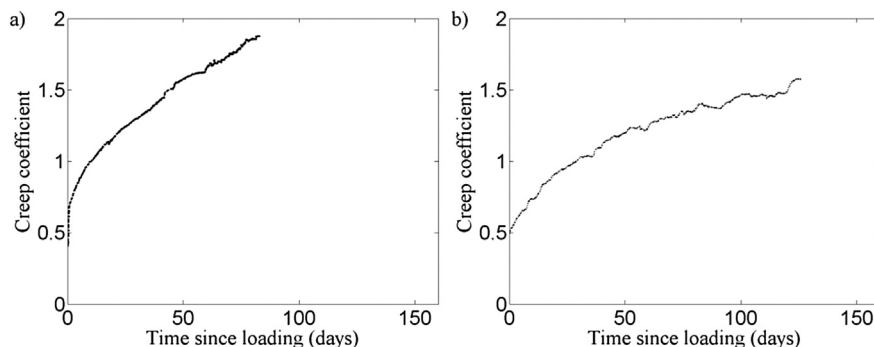


Fig. 5. Experimental creep coefficient of a) C1 series and b) C2 series.

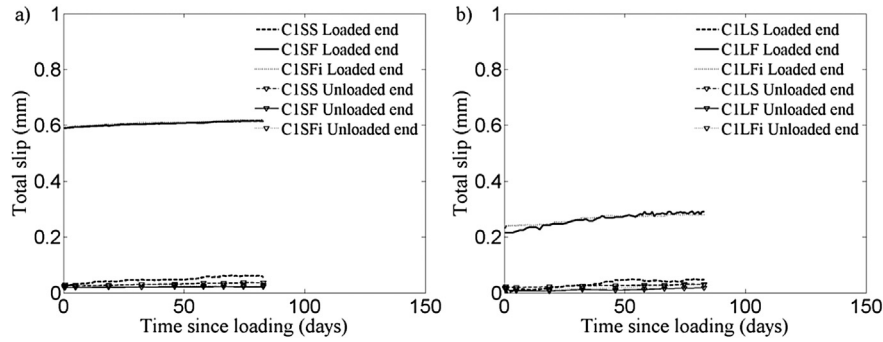


Fig. 6. Experimental total slip vs. time after loading for C1 specimens with a) short bond length and b) long bond length.

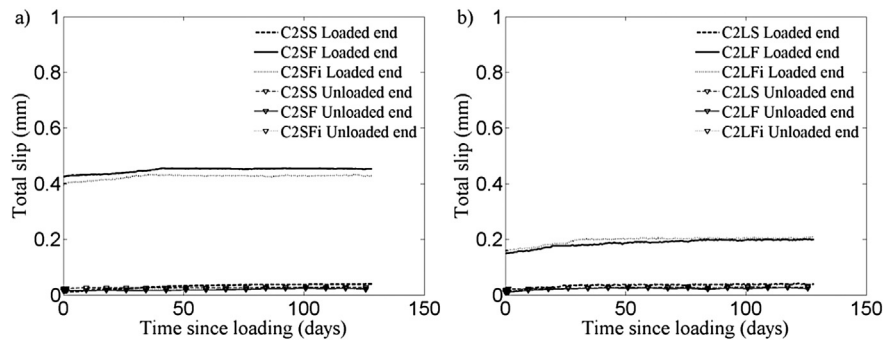


Fig. 7. Experimental total slip vs. time after loading for C2 specimens with a) short bond length and b) long bond length.

Table 7

Experimental results for immediate slip.

Specimen	Immediate slip (mm)	
	Loaded end	Unloaded end
C1SS	0.028	0.025
C1SF	0.590	0.019
C1SFi	0.591	0.021
C1LS	0.018	0.006
C1LF	0.216	0.014
C1LFi	0.233	0.015
C2SS	0.021	0.006
C2SF	0.424	0.016
C2SFi	0.401	0.015
C2LS	0.016	0.002
C2LF	0.148	0.010
C2LFi	0.158	0.009

As indicated previously, this experimental investigation included four pull-out specimens whose reinforcement was internally strain gauged so that data on the reinforcement strains along the bond length would be available. In this section, the reinforcement strain distributions are presented and analysed. In a further step, these strain distributions are used for the analysis of the bond profile and redistribution of bond stresses over time which take place due to sustained loading.

3.3.1. Reinforcement strain distribution

Fig. 10 shows the strain gauge readings at various locations along the bond and unbonded length (see Fig. 2b), with location 0 mm representing the unloaded end of the pull-out specimen. The figure includes two different sets of information. First, strain development with load is represented at the very first curves (plotted at 0 days after loading). These curves were registered during the instantaneous test, where load was progressively applied until 15% of the ultimate capacity of the GFRP reinforcing

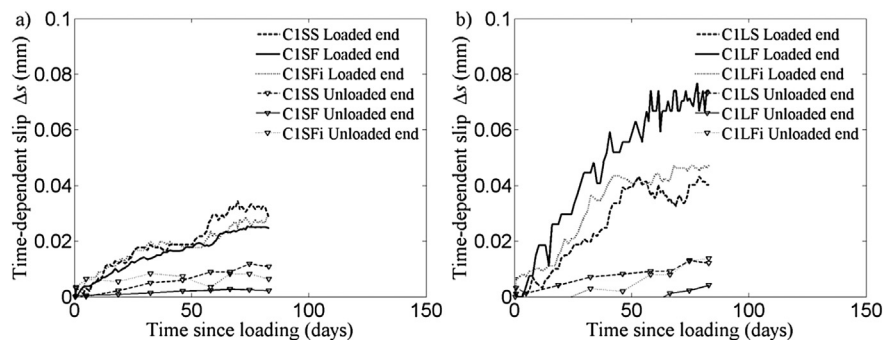


Fig. 8. Experimental time-dependent slip vs. time after loading for C1 specimens with a) short bond length and b) long bond length.

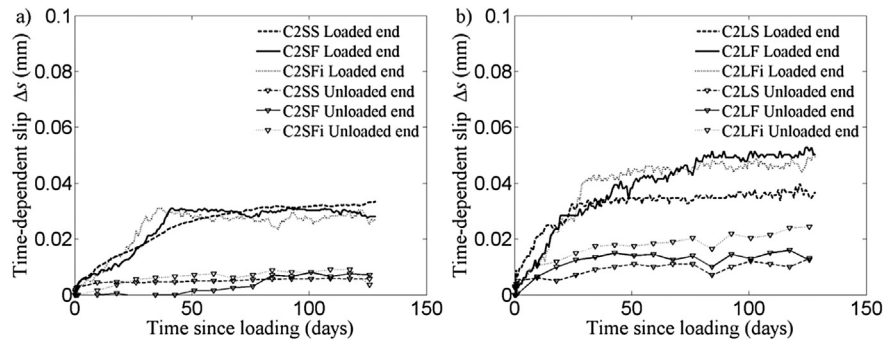


Fig. 9. Experimental time-dependent slip vs. time after loading for C2 specimens with a) short bond length and b) long bond length.

Table 8

Experimental time-dependent slip at different times during long-term testing.

Specimen	Δs_{60} (mm)		Δs_{90} (mm)		Δs_{130} (mm)	
	Loaded	Unloaded	Loaded	Unloaded	Loaded	Unloaded
C1SS	0.029	0.009	0.029	0.011	—	—
C1SF	0.023	0.002	0.024	0.002	—	—
C1SFi	0.022	0.007	0.030	0.006	—	—
C1LS	0.040	0.009	0.040	0.012	—	—
C1LF	0.074	0.003	0.074	0.004	—	—
C1LFi	0.044	0.008	0.047	0.010	—	—
C2SS	0.028	0.005	0.031	0.006	0.030	0.006
C2SF	0.030	0.002	0.030	0.007	0.031	0.008
C2SFi	0.026	0.008	0.027	0.008	0.030	0.009
C2LS	0.036	0.010	0.036	0.010	0.037	0.011
C2LF	0.043	0.014	0.049	0.014	0.050	0.013
C2LFi	0.046	0.018	0.046	0.022	0.048	0.021

bar was reached. Second, strain development with time is represented with the help of the third axis. These curves were registered during the long-term test, when load was sustained for 90 days in the case of the C1 series and 130 days for the C2 series. For practical reasons, only data regarding specimen C1LFi is shown in Fig. 10, although this information is also available for specimens C1SFi, C2SFi and C2LFi.

Fig. 11 shows the reinforcement strain distribution at different load levels at the time of loading (instantaneous pull-out corresponding to 0 days in Fig. 10), with location 0 mm representing the unloaded end of the pull-out specimen and the vertical dashed lines representing the location of strain gauges (see Fig. 2b). The

bond length is clearly marked to help in the analysis of the results. Some of the strain gauges failed during the test, and therefore no data is recorded at some locations. For example, the strain gauge at 104 mm from the unloaded end of specimen C1SFi failed, and this is evident in Fig. 11a, where no marker exists at that location. In this case, for illustrative purposes, and because this gauge was located outside the bond length, its value was assumed to be the same as that of location 280 mm, also located outside the bond length.

A significant difference can be observed between Fig. 11a and b. The loading process in the short bond length is characterized by a quasi linear distribution of strains (see Fig. 11a) along the bond length, whereas in the case of long bond length a clearer non-linear distribution is obtained (see Fig. 11b). According to these results, bond stresses are more evenly distributed in a short bond length, with the entire bond region contributing to the stress transfer process. On the other hand, for specimens with long bond length the transfer length increases with the applied load.

Reinforcement strain distribution at different times during long-term testing is shown in Table 9 and Figs. 12 and 13. As in previous figures, location 0 mm represents the unloaded end, vertical dashed lines represent the location of strain gauges and bond length is clearly marked.

Irrespective of the length of the bond zone and the concrete grade, experimental results show that no significant increase in the readings of the strain gauges located in the unbonded zone is visible at the service load levels applied in the present work (see section 2.3). This means that no appreciable creep due to sustained loading occurred in the GFRP bar as expected. In the region within the bond length, the effect of sustained load is reflected in an

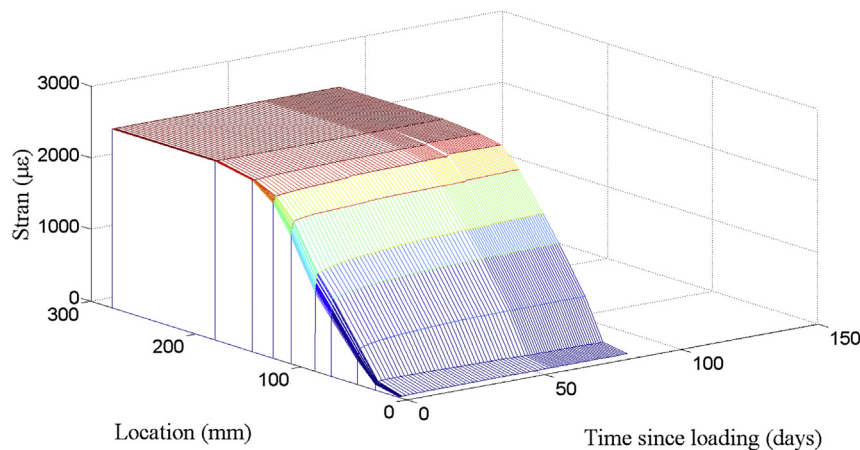


Fig. 10. Experimental reinforcement strain distribution over time for specimen C1LFi.

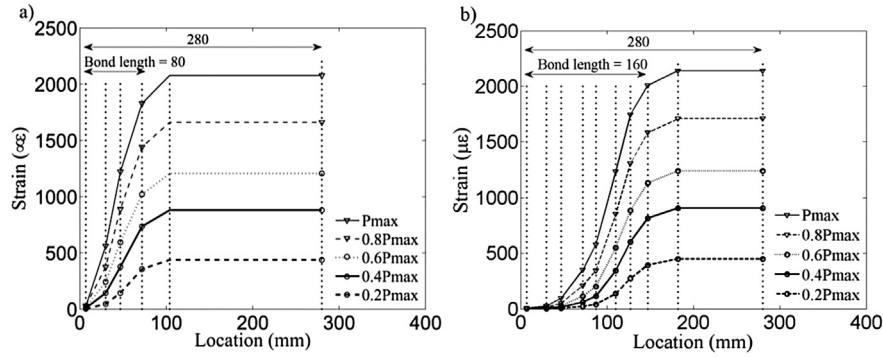


Fig. 11. Experimental reinforcement strain distribution at different loads during instantaneous pull-out test for specimen with a) short bond length (C1SFi) and b) long bond length (C1LFi).

increase in the strain gauge readings, with higher rates occurring at the intermediate zone between loaded and unloaded ends. This can be attributed to bond deterioration and redistribution of stresses, which is lower in more damaged zones (i.e. close to loaded end) and in zones with low levels of stress (i.e. unloaded end in specimens with larger bond length). This is indicated by the area between curves after loading and 90 days in Figs. 12 and 13. The stabilization of this redistribution occurs at approximately 30 days after loading in the case of the C1 series (see Fig. 12). For the C2 series, the increase in strains slows down and stabilizes at the end of long-term testing (see Fig. 13). In situations in which creep could not be ignored, higher deformations and therefore larger redistributions and slips would be expected.

Table 9
Experimental instantaneous strain and time-dependent strain at different times during long-term testing.

Specimen	Strain gauge	Position (mm)	Instantaneous strain, ϵ_{inst} ($\mu\epsilon$)	Strain at 60 days, ϵ_{60} ($\mu\epsilon$)	Strain at 90 days, ϵ_{90} ($\mu\epsilon$)	Strain at 130 days, ϵ_{130} ($\mu\epsilon$)
C1SFi	G1	7	63	150	276	—
	G2	30	948	1680	1702	—
	G3	47	1582	2070	2084	—
	G4	72	2123	2250	2243	—
	G5	104	Failed	—	—	—
	G6	280	2421	2422	2423	—
C1LFi	G1	7	8	30	31	—
	G2	30	73	90	87	—
	G3	47	307	660	645	—
	G4	72	731	1260	1270	—
	G5	87	1008	1590	1615	—
	G6	110	1688	2091	2121	—
	G7	127	2115	2370	2390	—
	G8	147	2385	2438	2449	—
	G9	182	2494	2496	2497	—
	G10	280	2493	2496	2495	—
C2SFi	G1	7	Failed	—	—	—
	G2	30	567	930	983	1077
	G3	47	1013	1417	1474	1563
	G4	72	1981	2260	2280	2319
	G5	104	Failed	—	—	—
	G6	280	Failed	—	—	—
C2LFi	G1	7	14	126	142	161
	G2	30	Failed	—	—	—
	G3	47	115	342	377	422
	G4	72	368	902	941	990
	G5	87	531	1189	1256	1319
	G6	110	930	1637	1702	1781
	G7	127	1233	1845	1916	2037
	G8	147	1711	2082	2239	2215
	G9	182	2269	2270	2272	2274
	G10	280	2272	2273	2273	2274

Reinforcement strains both at the beginning (ϵ_{inst}) and at 90 days (ϵ_{90}) of the sustained loading tests are greater for C1 series specimens, as might be expected. However, larger increments in strains due to long-term testing ($\Delta\epsilon_{90}$) are found for C2 series specimens with long bond lengths (see Table 9). This is a sign of greater stress transfer, which is related to better bond behaviour attributable to a higher concrete grade.

3.3.2. Bond stress distribution

The experimental data shown in the previous section was used to analyse bond deterioration due to sustained loading. Taking the definition of bond as the force transferred between the reinforcing bar and the concrete per unit of surface area, and performing equilibrium calculations on a piece of bar of length dx , along with the assumption of linear elastic behaviour of the bar, leads to:

$$\tau = \frac{EA}{\pi d_r} \frac{d\epsilon_r(x)}{dx} \quad (2)$$

where τ is the bond stress, EA is the axial stiffness of the reinforcement and d_r is the nominal bar diameter.

Eq. (2) was applied to experimental reinforcement strains registered during long-term testing, and experimental bond stress distributions were obtained for internally strain-gauged specimens (i.e. specimens C1SFi, C1LFi, C2SFi and C2LFi). Information regarding bond stress distributions for specimens of series C1 and C2 at different stages of long-term testing is presented in Figs. 14 and 15 respectively, with location 0 mm representing the unloaded end and as in previous figures. When Eq. (2) is applied between two consecutive strain gauges, $dx = \Delta x$, $d\epsilon_r = \Delta\epsilon_r$, and discrete constant values of bond stresses are obtained. It should be noted that bond stress distributions for specimen C2SFi is only partially represented due to the failure of one of the strain gauges (Fig. 15a).

Comparison of the plots of bond stresses at the time of application of sustained loading and at 90 days reveals a general trend of redistribution of bond stresses due to long-term loading. Within this redistribution, different trends can be found if short and long bond specimens are compared.

For short bond length specimens, bond stresses decrease at the loaded end and increase near the unloaded end, which is due to an increase of bar-concrete interface damage in the loaded end zone. This is a sign of the whole bond length contributing to the bond phenomenon. As an example, bond stress decreases from 5.9 MPa to 1.7 MPa at the loaded end and increases from 10.8 MPa to 18.5 MPa at the unloaded end in specimen C1SFi (see Fig. 14a), meaning a decrease of 70% and an increase of 71% respectively. Similar trends can be observed for specimen C2SFi (see Fig. 15a), with bond stress at the loaded end decreasing from 10.7 MPa to

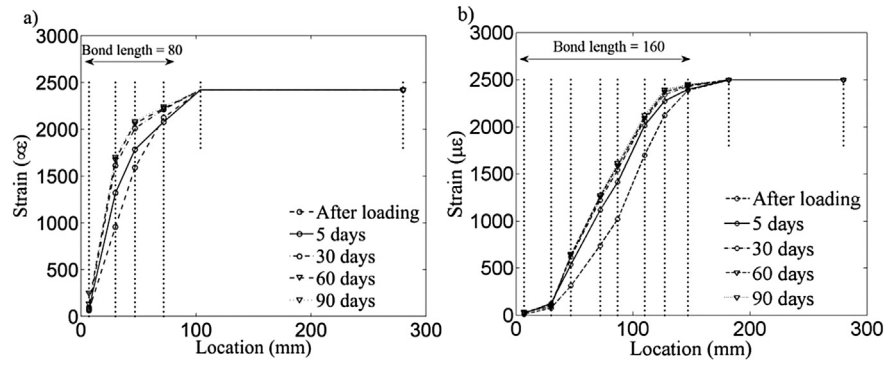


Fig. 12. Experimental reinforcement strain distribution at different times during long-term testing for specimen a) C1SF1 and b) C1LF1.

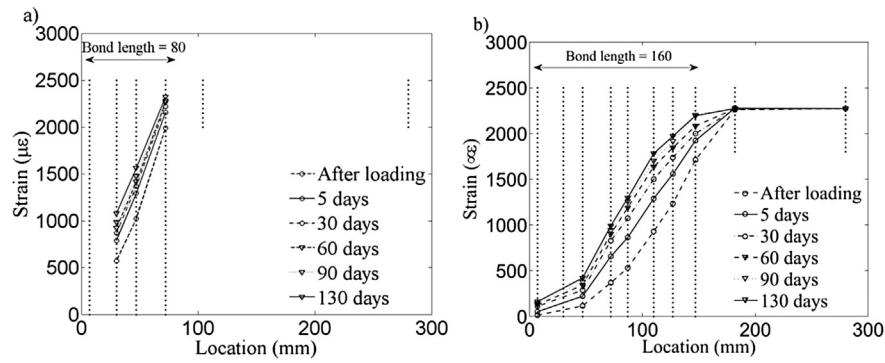


Fig. 13. Experimental reinforcement strain distribution at different times during long-term testing for specimen a) C2SF1 and b) C2LF1.

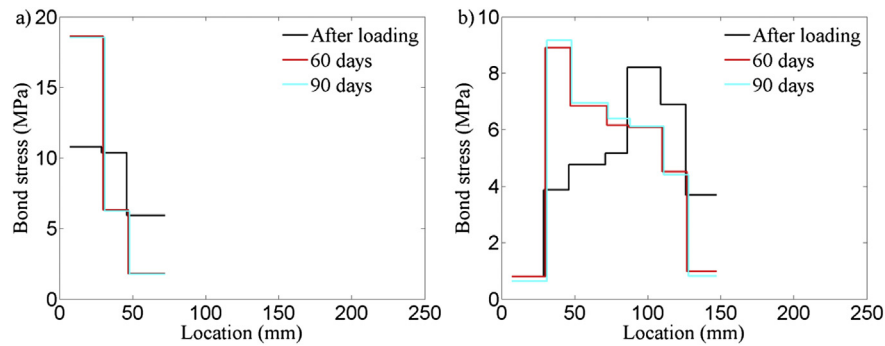


Fig. 14. Experimental bond stress distribution at different times during long-term testing for specimen a) C1SF1 and b) C1LF1.

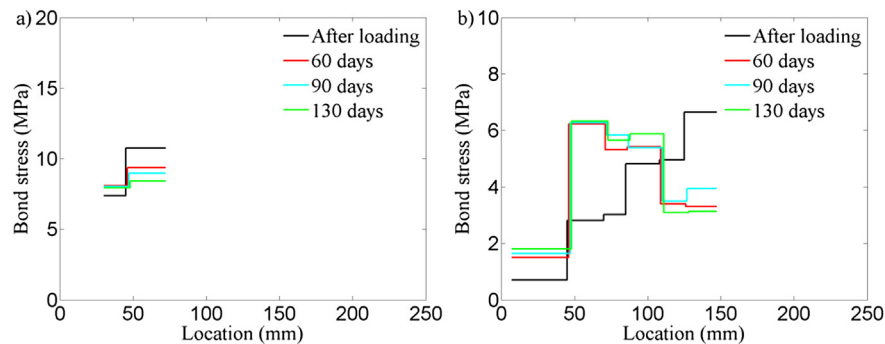


Fig. 15. Experimental bond stress distribution at different times during long-term testing for specimens a) C2SF1 and b) C2LF1.

8.9 MPa, a reduction of 16%. No percentage can be defined at the unloaded end due to the failure of one of the strain gauges.

In the case of long bond length specimens, a redistribution of bond stresses is also visible, with bond stresses decreasing at the loaded end, and larger bond stresses developing in neighbouring sections. As an example, bond stress at the loaded end decreases from 3.7 MPa to 0.8 MPa and from 6.6 MPa to 3.9 MPa in specimens C1LFI (see Fig. 14b) and C2LFI (see Fig. 15b) respectively, representing a reduction in bond stress of 78% and 41% respectively. However, such redistribution does not take place at the unloaded end of the specimens, indicating that the stress transfer process does not affect the entire length of the bond in the same way.

4. Conclusions

The results of an experimental programme to investigate the bond stress-slip behaviour of GFRP bars in concrete under sustained load have been presented and discussed. A total of 12 pull-out specimens with two concrete strengths (35 and 50 MPa), two bond lengths ($5d_f$ and $10d_f$) and two reinforcement materials (GFRP and steel) were tested. The specimens were initially subject to short-term loading, which was sustained for a period of between 90 and 130 days. A load corresponding to 15% of the ultimate capacity of the GFRP bar was selected so as to correspond to the service range. Based on the results of this study, the following conclusion can be drawn:

- Experimental results on immediate slip due to short-term loading confirm results from previous studies with larger slips for specimens with either lower concrete compressive strength or shorter bond length, attributable to larger local deformability and damage in the concrete surrounding the bar. At the same time the tendency for GFRP pull-out specimens to bars to have larger slips than steel specimens is also confirmed, which may attributed to the deformability jointly with the mechanical interlock characteristics of the bars. Globally, the differences in slips between both ends indicate a non-uniform distribution of stresses along the bonded length, which is more pronounced for larger bond lengths.
- Data from the internal strain gauges during short-term testing shows that bond stresses are more evenly distributed in a short bond length where a quasi linear distribution of strains can be observed. This is not the case in a long bond length, where distribution is more clearly non-linear.
- As regards time-dependent slip due to sustained loading, results show an increase in the slip that stabilizes at approximately 60 days after loading, irrespective of the concrete strength, with higher increments for long bond length specimens. This can be attributed to bond deterioration and redistribution of stresses, which is more pronounced at the intermediate zone between loaded and unloaded ends. The experimental results confirm the influence of concrete strength on the deterioration of bond performance, with higher concrete compressive strength showing a time-dependent slip around 23.3% lower than that of lower concrete compressive strength.
- The evolution of bond stresses under sustained load was analysed using data from internal strain gauges. The experimental results confirm the influence of concrete strength on the stress transfer process and the redistribution of stresses along the bond length, decreasing in neighbouring loaded end zones due to an increase of bar-concrete interface damage. Specimens with the greatest concrete compressive strength showed a smaller loss of bond stresses at the loaded end, with values of 16% and 41% (for short and long bond length respectively) compared to 70% and 78% for the lowest concrete strength. In addition,

irrespective of the concrete compressive strength, bigger variations of bond stresses were found in long bond length specimens, a sign of greater bond stress redistribution along the bond length.

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